

Analysis of Architectural Design Choices for Brain-to-Text Decoding

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- ① Introduction
- ② Data
- ③ Architecture
 - Day Alignment
 - Temporal Compression
 - Gated Recurrent Unit
 - Linear Classifier
- ④ Conclusion



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- Motivation: Restore communication for individuals who have lost the ability to speak due to conditions like Amyotrophic Lateral Sclerosis (ALS) or brainstem strokes
- Solution: Translate high-dimensional neural activity from the motor cortex directly into a sequence of phonemes or text

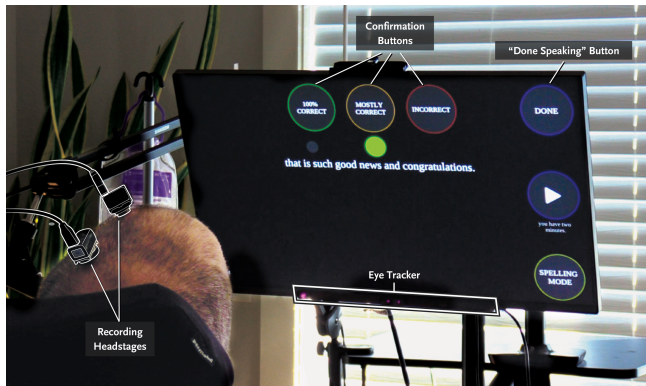


Figure 1: Conversation-Mode User Interface. (N. S. Card et al. 2024)

- Researchers from UC Davis developed a Brain-Computer Interface (BCI) to decode intracortical neural data using deep learning
 - Participant: 45-year-old male with ALS and severe dysarthria
 - Translated neural signals into text, achieving a remarkable 97.5% overall accuracy (2.5% Word Error Rate)

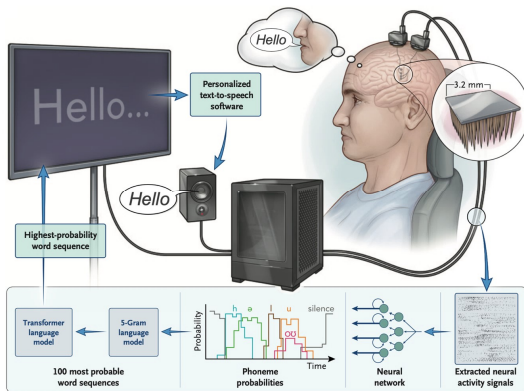


Figure 2: brain-to-text decoding pipeline (N. S. Card et al. 2024)



Session: t15.2023.09.01, Block: 9, Trial: 20

Sentence label: He's just goofing off like he always has.

True sequence: HH IY Z | JH AH S T | G UN F IH NG | AO F | L AY K | HH IY | AO L W EY Z | HH AE Z |

Predicted Sequence: HH IY Z | JH AH S T | G UN V IH NG | AO F | L AY K | HH IY | AO L W EY Z | HH AE Z |

Phoneme Error rate: 2.86%

Session: t15.2023.09.01, Block: 8, Trial: 6

Sentence label: That's what they said.

True sequence: DH AE T S | W AH T | DH EY | S EH D |

Predicted Sequence: DH AE T S | W AH T | DH EY | S EH D |

Phoneme Error rate: 0.00%

Session: t15.2023.09.01, Block: 9, Trial: 19

Sentence label: They live with us.

True sequence: DH EY | L AY V | W IH DH | AH S |

Predicted Sequence: DH EY | L AY V | W IH DH | AH S |

Phoneme Error rate: 0.00%

Session: t15.2023.09.01, Block: 8, Trial: 17

Sentence label: She didn't announce that to you.

True sequence: SH IY | D IH D AH N T | AH N AW N S | DH AE T | T UW | Y UW |

Predicted Sequence: SH IY | D IH D AH N T | AH N AW N T S | DH AE T | T UW | Y UW |

Phoneme Error rate: 3.85%

Session: t15.2023.09.01, Block: 9, Trial: 11

Sentence label: My name is Pat Johnson and I live in Texas.

True sequence: M AY | N EY M | IH Z | P AE T | JH AA N S AH N | AH N D | AY | L AY V | IH N | T EH K S AH S |

Predicted Sequence: M AY | N EY M | IH Z | P AE T | CH AH N S AH N | AH N D | AY | L AY V | IH N | T EH K S AH S |

Phoneme Error rate: 4.88%

Figure 3: Predicted phonemes using neural network(N. S. Card et al. 2024)



- UC Davis Study showed promising results for medical use, but:
 - No internal logic of why their model is effective
 - Finding the internal logic can reveal underlying biological patterns
- **Question: What makes their architecture successful?**
- Approach: analyze individual components of the phonetic decoder part of the entire architecture, isolating the impact of:
 - Data processing
 - Hyperparameters



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- Neural recordings from a 45-year-old male with ALS, spanning 45 sessions and comprising 10,948 spoken sentences
 - Each session contains around 250 trials (sentences)
 - One sentence = One trial
- A trial contains:
 - One ground truth phoneme sequence and true sentence label
 - Phoneme: Set of the smallest phonetic units to distinguish words

Sentence label: That's what they said.
True sequence: DH AE T S | W AH T | DH EY | S EH D |

Figure 4: Ground truth phoneme sequence and true sentence label (N. S. Card et al. 2024)

- Neural tensor with dimensions $D \times T$; T-time steps, D-dimensions

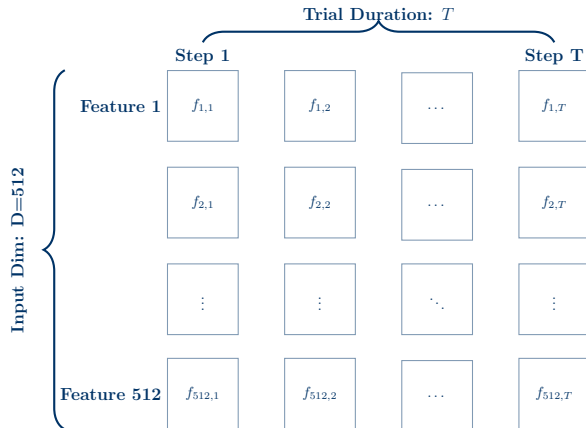


Figure 5: Structural representation of a single trial's neural data $D \times T$



Batching - 64 trials per batch

- Sampled from 4 distinct sessions with replacement (16 trials each)
- Prevents overfitting to non-stationary session-specific distributions

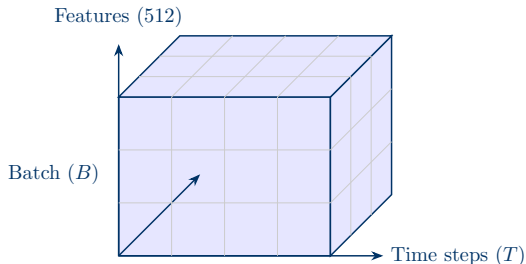


Figure 6: B trials in the batch, T time steps in a trial, 512 neural features in a time step ($\mathbb{R}^{B \times T \times 512}$)



Phoneme Error Rate (PER)

$$PER = \frac{S+I+D}{N}$$

True Label ($N = 6$): S P E E C H

Prediction: P E A C H Y

- **S:** 1 ($E \rightarrow A$ substitution)
- **I:** 1 (Y inserted at end)
- **D:** 1 (S deleted from start)

Calculation : $\frac{1+1+1}{6} = 0.50$

- **Levenshtein Distance:**
Measures the minimum edits required to align the prediction with the ground truth label
- Lower PER indicates higher decoding accuracy

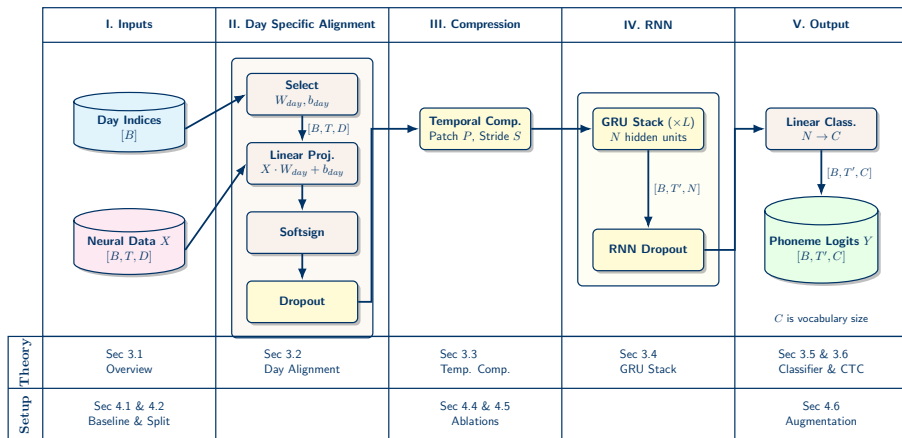
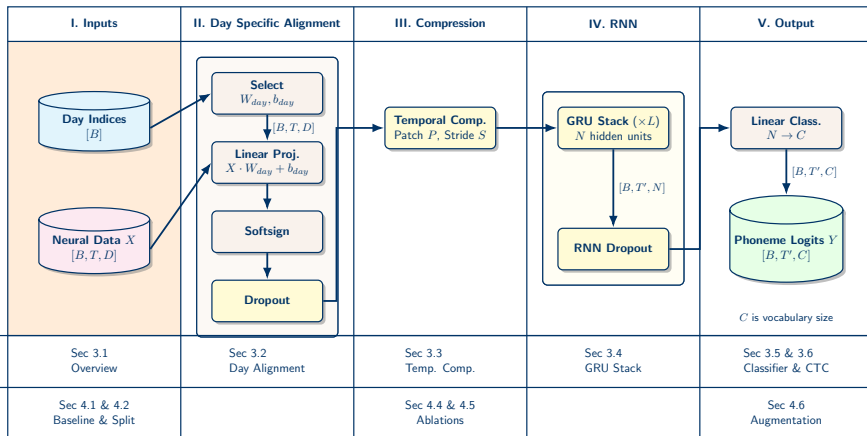


Figure 7: The pipeline for brain-to-text decoding. The architecture processes raw neural data X and session indices through linear day alignment, temporal compression, a multilayer GRU, and a final linear classifier to produce phoneme logits Y





Data: $X \in \mathbb{R}^{B \times T \times D}$

Neural data is inherently noisy, requiring stochastic transformations during training to improve generalization

- Additive White Noise: Injecting Gaussian noise (std 1.0) into the input tensor ($[B, T, D]$)
- Constant Offset Noise: Adding a broadcasted constant noise (std 0.2) to simulate natural baseline shifts ($[B, 1, D]$)
- Random Cut: Removing up to 3 time steps at the beginning of a trial to vary the speech start time
- Gaussian Smoothing: Applying a 1D Gaussian kernel along the time axis to reduce erratic spikes



Config.	Modification	PER (%)	Δ (%)
Baseline	All Aug.	10.54%	0.00%
Aug	No Aug.	18.32%	+7.78%
Aug	Constant Offset Only	17.74%	+7.20%
Aug	Gaussian Smooth Only	16.76%	+6.22%
Aug	Random Cut Only	16.19%	+5.65%
Aug	White Noise Only	12.20%	+1.66%

- White noise: the most impactful individual augmentation; a powerful regularizer that lowered the PER to 12.20%
- Constant offset noise and random cutting yielded minor improvements, but are useful for maintaining stability against non-stationary feature means and timing variations



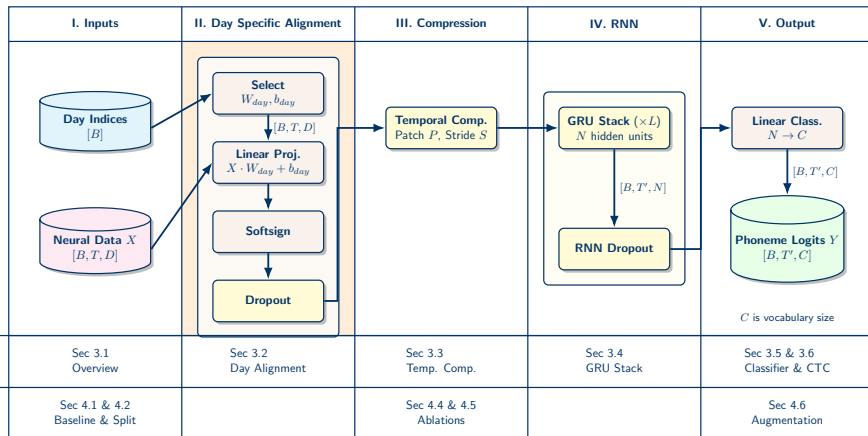
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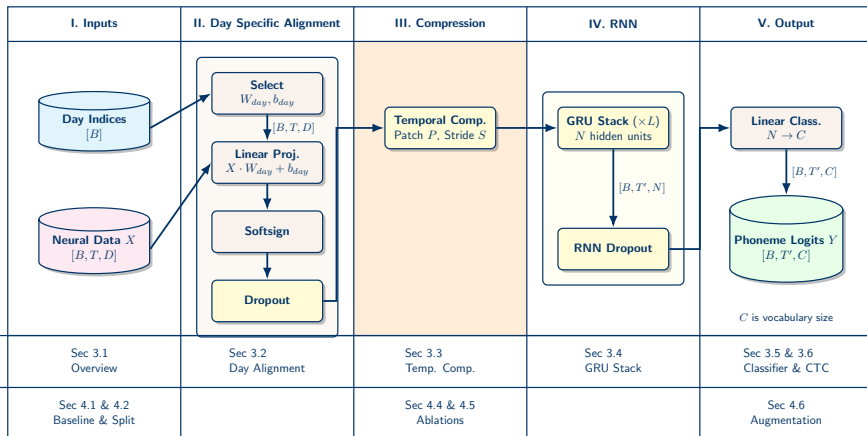
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Input: $X \in \mathbb{R}^{B \times T \times D}$

- Correcting for non-stationarity between different sessions because of microshifts in the electrode array placement
 - $Z = X \cdot W_{day} + b_{day}$
 - $W_{day} \in \mathbb{R}^{45 \times D \times D}$, $b_{day} \in \mathbb{R}^{45 \times D}$ for 45 days
- A Softsign activation to prevent exploding gradients
 - $A = \frac{Z}{1+|Z|}$
- An inverted dropout layer to prevent over-reliance on single features
 - $M_{b,t,d} \sim \text{Bernoulli}(1 - p)$
 - Output: $\tilde{X} = (A \odot M) \frac{1}{1-p}$
 - $\tilde{X} \in \mathbb{R}^{B \times T \times D}$





- A sliding window operation applied across the temporal dimension, defined by a **Patch size (P)** and a **Stride (S)**
- Flattens consecutive time steps into a compressed, high-dimensional vector
- Reduces the overall sequence length that the recurrent layers must process
 - $T' = \lfloor \frac{T-P}{S} \rfloor + 1$
 - $D' = P \times D$
 - $U \in \mathbb{R}^{B \times T' \times D'}$

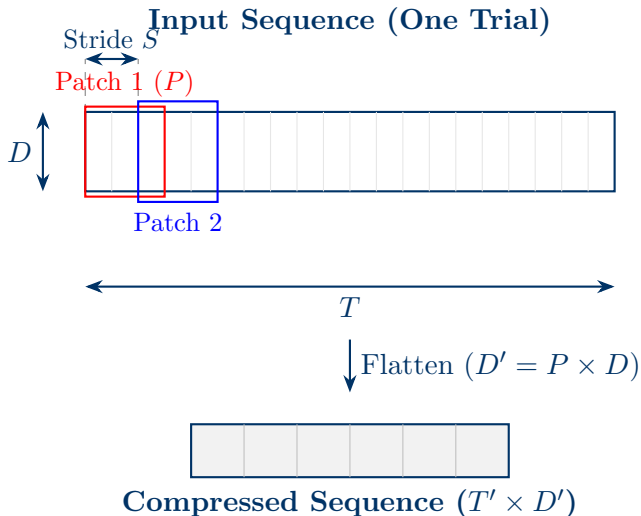


Figure 8: Mechanism of temporal patching



Patch Size	Stride 1	Stride 2	Stride 3	Stride 4
22	–	–	–	10.42% (–0.12%)
14	–	10.54% (0.00%)	10.59% (+0.05%)	10.54% (Baseline)
10	–	–	–	10.97% (+0.43%)
6	–	–	–	11.17% (+0.63%)
2	10.77% (+0.23%)	–	–	–
1	22.04% (+11.50%)*	–	–	–

*Patch 1, Stride 1 is equivalent to Raw Sequences (No Temporal Compression).

Figure 9: Effects of patches and strides on the convergent validation PER (%)

- Baseline: Patch Size (P) = 14, Stride (S) = 4
- Removing the patching mechanism doubled the PER from 10.54% to 22.04%
- Even a minimal patch size of 2 adjacent frames resulted in a nearly 12% reduction in validation PER compared to no patching

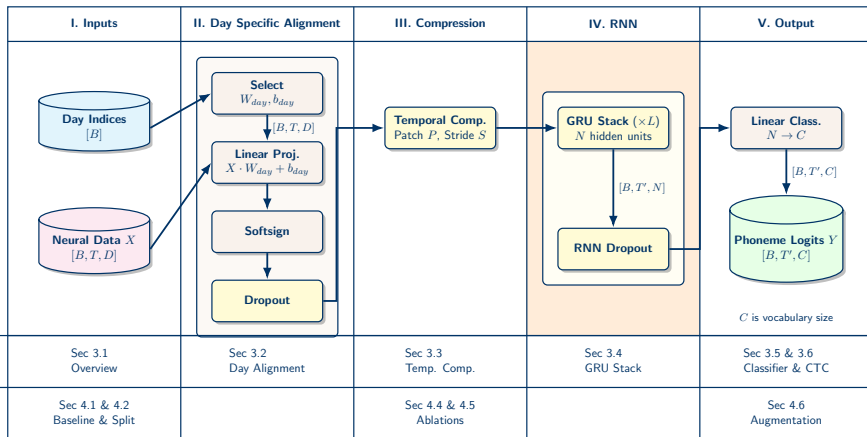


- Increasing patch size generally lowers PER, though improvements become marginal past a size of 14 time steps
- Stride size had minimal impact on accuracy, meaning a larger stride is preferred to reduce computational overhead

Purpose:

Allows the network to learn the underlying temporal trajectories and “motion” of neural signals instead of modeling micro-fluctuations between isolated frames (Boullard and Morgan 1994)

- Captures long-range dependencies and high-level phonetic logic within the neural signals
- Overlaps between the patching windows improve decoding performance





- A multilayer unidirectional GRU processes the compressed temporal features
- Sequentially updates a hidden state vector to encode the temporal history and linguistic intent of the trial
- Modulated by an **update gate** (determines how much past memory to carry forward) and a **reset gate** (determines how much to forget)
- $U \in \mathbb{R}^{B \times T' \times D'}$
- Output: $H^{(n)} \in \mathbb{R}^{B \times T' \times N}$, N : number of hidden units

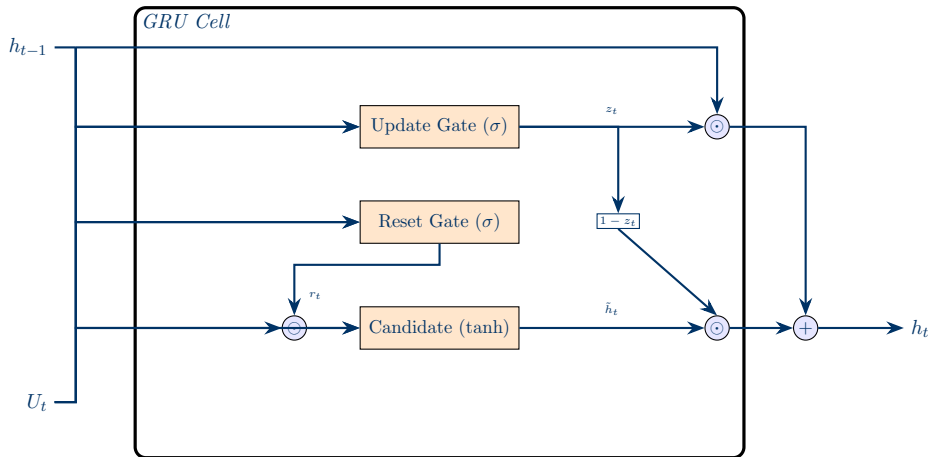
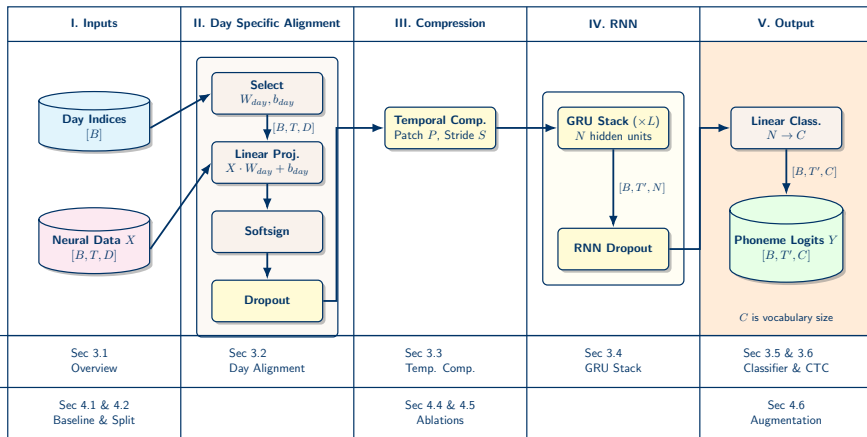


Figure 10: The update gate z_t and reset gate r_t modulate the flow of information from the previous hidden state h_{t-1} and the current compressed neural input U_t to produce the new state h_t .



Config.	Modification	PER (%)	Δ (%)
Baseline	Baseline	10.54%	0.00%
Depth	GRU (2 Layers)	12.19%	+1.65%
Depth	GRU (1 Layer)	16.68%	+6.14%
Width	512 Hidden Units	11.14%	+0.60%
Width	1024 Hidden Units	10.38%	-0.16%

- Baseline: 5 stacked GRU, 768 hidden units, RNN dropout: 0.4
- Model Depth: A multi-layer hierarchy is essential for extracting abstract linguistic understanding
- Model Width: Modifying the representational capacity also impacted performance





- A dense feedforward layer maps the latent GRU representations into unnormalized logits over a 41-token phoneme vocabulary
 - Input: $H^{(n)} \in \mathbb{R}^{B \times T' \times N}$
 - $y_t = W_{out}h_t + b_{out}$. W_{out} and b_{out} shared by all time steps and trials
 - Output: $Y \in \mathbb{R}^{B \times T' \times C}$, C : number of phonemes (41)
- Logits will be turned into probability distributions for each time step via SoftMax



Connectionist Temporal Classification loss:

- No alignment between neural signals and discrete phonemes
- CTC maximizes the probability of producing neural-to-phoneme alignments that could produce the target sequence

Ex: 280 (14 time steps) of neural activity produce the word “HELLO”

- Alignment 1: – – HH – -E LL OW – – → “HELLO”
- Alignment 2: HH EE L L OW – – – – – → “HELLO”
- Alignment 3: - HH - EE - L - L - OW - - → “HELLO”
- Alignment 4: HH HH EL L - L OW OW - - - → “HELLO”



CTC Example - 2 Time Steps (- -): A is the true sequence

Probability Distribution Table

Token	$t = 1$	$t = 2$
$P(-)$	0.5	0.4
$P(A)$	0.4	0.3
$P(B)$	0.1	0.3

- Many-to-One Mapping: Marginalizes over all valid alignments to find the most likely sequence

Sequence: “A”

$$P(\text{“A-”}) = 0.4 \times 0.4 = 0.16$$

$$P(\text{“-A”}) = 0.5 \times 0.3 = 0.15$$

$$P(\text{“AA”}) = 0.4 \times 0.3 = 0.12$$

$$P(S = \text{“A”}) = 0.16 + 0.15 + 0.12 = 0.43$$

$$P(S = \text{“B”}) = 0.22, P(S = \text{“”}) = 0.20,$$

$$P(S = \text{“AB”}) = 0.12, P(S = \text{“BA”}) = 0.03$$

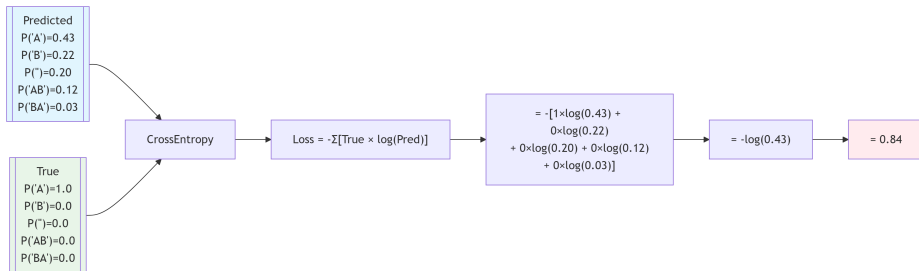


Figure 11: CTC loss calculation with the path probabilities



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Final Model Parameters:

Patch Size: 22, Stride: 4, Full Data Augmentations, 1024 hidden units, and all regularizations (dropout)

- Validation Phoneme Error Rate: 10.25%
- Final Testing Phoneme Error Rate: 9.68%



- Key Insight: Temporal compression is the most critical architectural component, allowing the model to focus on the trajectory of underlying neural motion
- Future Directions
 - Integration of attention mechanisms/Transformers
 - Developing transfer learning frameworks to generalize beyond a single clinical participant
 - Test the hypothesis that Temporal Compression allows: the network to learn the underlying “motions” of the neural signals, offloads the burden of learning local fluctuations
 - Overlaps between the patching windows improve decoding performance

Thank you for your attention!
Questions?



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 - Each session contains around 250 trials (sentences)
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 - Phoneme: Set of the smallest phonetic units to distinguish words

Sentence label:	That's what they said.
True sequence:	DH AE T S W AH T DH EY S EH D

Figure 12: Ground truth phoneme sequence and true sentence label (N. S. Card et al. 2024)

- Neural tensor with dimensions $D \times T$; T-time steps, D-dimensions
 - 256 (-4.5 RMS) threshold crossings (neurons firing)
 - 256 spike-band power values (average power of electrical signal)
 - Rolling Z-score normalization for each feature with the past 20 trials
 - Gaussian smoothing with a standard deviation of 2 time steps



Compute Infrastructure

- UNC Longleaf HPC: NVIDIA Volta GPU + 8 CPU cores
- State Dictionary for optimizer state, learning rate scheduler state, and model parameters (continue training)

Batching - 64 trials per batch

- Sampled from 4 distinct sessions with replacement (16 trials each)
- Prevents overfitting to non-stationary session-specific distributions

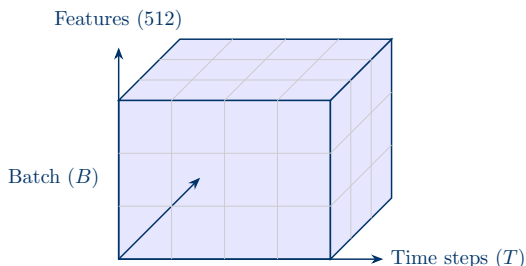


Figure 13: B trials in the batch, T time steps in a trial, 512 neural features in a time step ($\mathbb{R}^{B \times T \times 512}$)



Optimization

- Adam Optimizer with a weight decay of 0.001
- Learning Rate Schedule
 - 1,000 step linear warm up to 0.005
 - Cosine decay to 0.0001 over 120,000 batches of training
- Optimization progress is measured in total batches processed
 - Distributional balance: find common signals between the 4 sessions
 - Validation occurs every 2000 batches trained
 - A best checkpoint of model weights is saved if the validation error rate reaches a new minimum